

CSC360 - Computer Graphics and Digital Image Processing

10/9/26 Reflection

Topic: Files, Streams, XML, Networking and Statistical Concepts

In today's lecture, I was able to learn about the differences between files and streams as well as their relationship with graphics, videos and statistical concepts. There was also a discussion on how XML and networking play roles in graphics.

Files and Streams

There is an exact quantity of stored data in a file as well as an exact point of start and end. But in the case of a stream, it can be considered as a continuous flow of data which cannot be determined in advance by the program where it ends.

- File → stored data of fixed nature with an exact size.
- Stream → continuous data flow.
- Streams are effective whenever data is received or processed in a continuous manner.
- A program can read data from streams in a gradual manner.

Streaming of Graphical Data and Video

In graphics, there is need for streaming where graphical data is required to be received and processed continuously. This way, there is no need for processing of all data that will be loaded.

In video, the need for streaming becomes essential because videos consist of continuous frames of images. The user can view the video even as the rest of the video is receiving.

Difference Between Population Mean and Sample Mean

We further examined the difference between population mean and sample mean.

- Population mean → mean of all values in the total population.
- Sample mean → mean of values in a selected sample of the population.
- Sample mean is used when analyzing the total population is hard or expensive.

Expected Value of X

The expected value of X, denoted by $E(X)$, is the expected value or average of a random variable.

If X is a discrete random variable:

$$E(X) = \sum x P(X = x)$$

The concept of probability ties in with that of an average.

Streaming and Expected Value

It was very interesting to learn about the application of streaming to the estimation of the expected value of X. In case of data arriving sequentially, we can update the average with each observation rather than keeping the entire sequence.

This becomes very helpful when the size of the data is large or even infinite.

XML and Graphics

In addition to this, we learned that XML may also be related to graphics, especially Scalable Vector Graphics (SVG).

- XML offers a means of describing data in a structured way.
- SVG makes use of XML elements to describe graphical elements.
- These include lines, circles, rectangles, and paths which can be described with XML tags.
- SVG drawing tools can create or alter these graphical elements.

This gave me an idea that graphics can be depicted in the form of structured text instead of being limited to pixels.

Graphics and Networking

Networks play an important role in graphics when there is a need to transfer graphical information between two computers.

Server-Side Computing versus Client-Side Computing

Server-side computing takes place on the server side and the output is sent to the user after that.

- Centralized computing.
- Computing is very powerful.
- Easy to manage and update.
- Requires a connection with the network.
- Increases server workload and reaction time.

Client-side computing takes place at the client's end.

- Faster interaction with data once acquired.
- Decreases the load of computing on the server.
- Doesn't require continuous server connection for some computations.
- Depends on the client's computing power.
- Updates and computations need to be carried out on multiple clients.

Key Takeaway

Streams are helpful in handling continuously arriving data, XML may be used to describe graphical objects like SVG figures, and networking facilitates transfer of graphical data from client to server. These are some of the examples that prove that computer graphics is not just about creating images but it is related to data handling as well.

